

A Causal Evaluation of the Impact of Motorcycles on the Severity of Traffic Near-Misses Using Causal Machine Learning and Massive Drone Trajectory Data

Abstract

The management of mixed traffic flows remains a fundamental challenge in traffic engineering due to the heterogeneous dynamic characteristics of vehicles. Previous studies have heavily relied on correlation-based statistical models, which face severe limitations in isolating causal effects from confounding variables. This study develops a Causal Machine Learning framework based on the Causal Forest algorithm to evaluate the isolated impact of Power-Two-Wheelers (PTWs) on the "Time-to-Collision" (TTC_{2D}) safety surrogate measure. To this end, massive drone-based traffic trajectory data (pNEUMA) comprising over 1.5 million movement records was utilized. To eliminate sensor noise and extract true physical near-misses, a continuous rolling window filter (25 consecutive frames, equivalent to 1 second) was applied. Furthermore, to overcome the "Perfect Separation" phenomenon in kinematic variables, Mahalanobis Distance Matching was employed. The results revealed that after achieving complete covariate balance, the presence of a motorcycle significantly reduces the TTC by an average of 0.1026 seconds, dropping the mean TTC in critical conditions from 0.39 to 0.25 seconds. Additionally, Rosenbaum Bounds sensitivity analysis proved that this causal inference is highly robust against unobserved confounders up to $\Gamma = 3.0$ and remains statistically significant ($p = 0.0059$).

Keywords: Traffic Safety, 2D Time-to-Collision, Causal Inference, Causal Forest, Mahalanobis Matching, Rosenbaum Bounds.

1. Introduction

Ensuring the safety of urban road networks and reducing crash rates has consistently been a primary concern in transportation engineering. In dense urban environments, the presence of Power-Two-Wheelers (PTWs) significantly complicates the dynamics of mixed traffic flows. Due to their smaller physical dimensions, rapid acceleration capabilities, and aggressive driving behaviors such as lane-splitting, motorcycles not only face higher collision risks but also introduce turbulence into the behavior of adjacent drivers (Vlahogianni et al., 2012).

Traditionally, traffic safety analysis has been conducted using historical crash records. However, the rare, random, and frequently under-reported nature of these events has driven researchers toward Surrogate Safety Measures (SSM) (Tarko, 2018). Among these, the "Time-to-Collision" (TTC) index is widely accepted as the most reliable indicator for assessing the severity of traffic conflicts.

The primary challenge in analyzing the TTC index lies in disentangling the true, inherent effect of a specific factor (e.g., the presence of a motorcycle) from other kinematic variables, such as higher relative speeds. Moving beyond traditional correlation approaches and adopting "Causal Inference" models is inevitable for robust microscopic traffic analysis. This research addresses this gap by deploying a Causal Machine Learning pipeline on high-resolution trajectory big data to evaluate the isolated physical impact of motorcycles on traffic safety margins.

2. Literature Review

A substantial portion of safety studies has utilized statistical regression models, such as Logit and Probit, to identify factors influencing crash severity. With the advent of massive trajectory data and drone-based databases like pNEUMA, it became possible to extract interactions at the millisecond level (Barmounakis & Geroliminis, 2020).

Despite these advancements, microscopic analyses often suffer from "Selection Bias" and "Perfect Separation." Previous studies have struggled to fully control for confounding variables, such as initial differences in relative speed, when comparing PTWs with passenger cars, especially without the interference of sensor tracking noise. This research bridges this methodological gap by combining massive trajectory data, continuous temporal filters (Run-Length Encoding), the novel "Causal Forest" approach, and Mahalanobis distance matching to provide a mathematically and physically rigorous causal evaluation.

3. Methodology

3.1. Database Description and Big Data Preparation

The data utilized in this study was extracted from the pNEUMA traffic database, which includes precise vehicle trajectories in the urban network of Athens with a temporal resolution of 0.04 seconds. To implement causal inference models while maintaining time-series integrity, over 1.5

million trajectory records were processed to extract interacting vehicle pairs (Car-Car as the control group and Car-PTW as the treatment group) frame-by-frame.

3.2. 2D Kinematics and Pure Near-Miss Extraction

Given the lateral movement characteristics of motorcycles, one-dimensional calculations lack physical validity. Since the data possessed Cartesian coordinates (UTM), the physical distance between two vehicles was calculated using the 2D Euclidean formula:

$$d = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2}$$

Furthermore, the 2D Time-to-Collision (TTC_{2D}) was computed using the dot product of the relative velocity ($\vec{v}$) and relative position ($\vec{r}$) vectors. This index was only considered valid under conditions of physical convergence ($\vec{r} \cdot \vec{v} > 0$):

$$TTC_{2D} = \frac{-(\vec{r} \cdot \vec{v})}{\|\vec{v}\|^2}$$

To completely eliminate sensor tracking noise and identify absolute near-misses, a rolling window filter was applied. Accordingly, interactions were only retained if the critical condition ($TTC \leq 3s$ and $Distance \leq 30m$) persisted continuously for at least 25 frames (1 full second). The application of this strict filter resulted in the extraction of 91 pure, noise-free critical events for each group.

3.3. Causal Machine Learning (Causal Forest) and Mahalanobis Matching

To estimate the Average Treatment Effect (ATE), the advanced Generalized Random Forests (Causal Forest) algorithm was utilized. The treatment variable (W) was the presence of a motorcycle, the outcome variable (Y) was the minimum TTC_{2D} , and the confounders (X) included initial distance, relative speed, and acceleration rate.

Additionally, to overcome the bias of heterogeneous distributions and enable sensitivity testing, "Mahalanobis Distance Matching" was employed to extract interacting pairs with strictly equivalent kinematic conditions.

4. Results

4.1. Causal Inference and CATE Analysis

Following the processing of 1.5 million records and kinematic isolation, descriptive statistics of the critical traffic events were extracted. The results are summarized in Table 1:

Table 1. Descriptive statistics of kinematic variables for pure critical near-miss events (N=91 per group).

Interaction Type	N (Events)	Mean TTC (s)	SD TTC (s)	Mean Relative Speed (m/s)	SD Relative Speed (m/s)	Mean Initial Distance (m)
Car - Car (Control)	91	0.39	0.4	5.34	1.61	12.82
Motorcycle - Car (Treatment)	91	0.25	0.27	5.8	1.76	13.13

The results of the Causal Forest algorithm demonstrated that the presence of a motorcycle, under similar kinematic conditions, exerts a negative causal effect on the TTC index. The model estimates that PTWs reduce the safety margin by an average of **-0.1026 seconds** (ATE = -0.1026s). Feature importance analysis within the algorithm revealed that initial distance (37.0%) and relative speed (33.0%) play the most significant roles in determining conflict severity.

The boxplots (Figure 1) and the distribution of Conditional Average Treatment Effects (Figure 2) illustrate the behavioral differences in safety margins and the heterogeneous impact of the treatment, respectively.

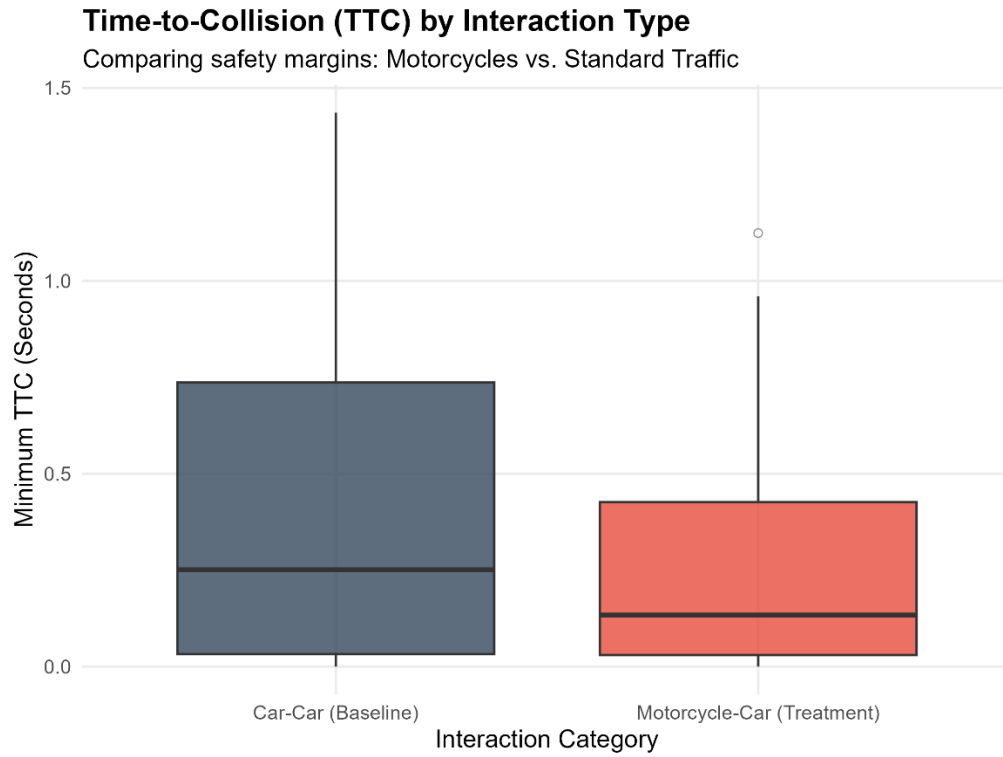


Figure 1. Distribution of Time-to-Collision (TTC) values in isolated critical events.

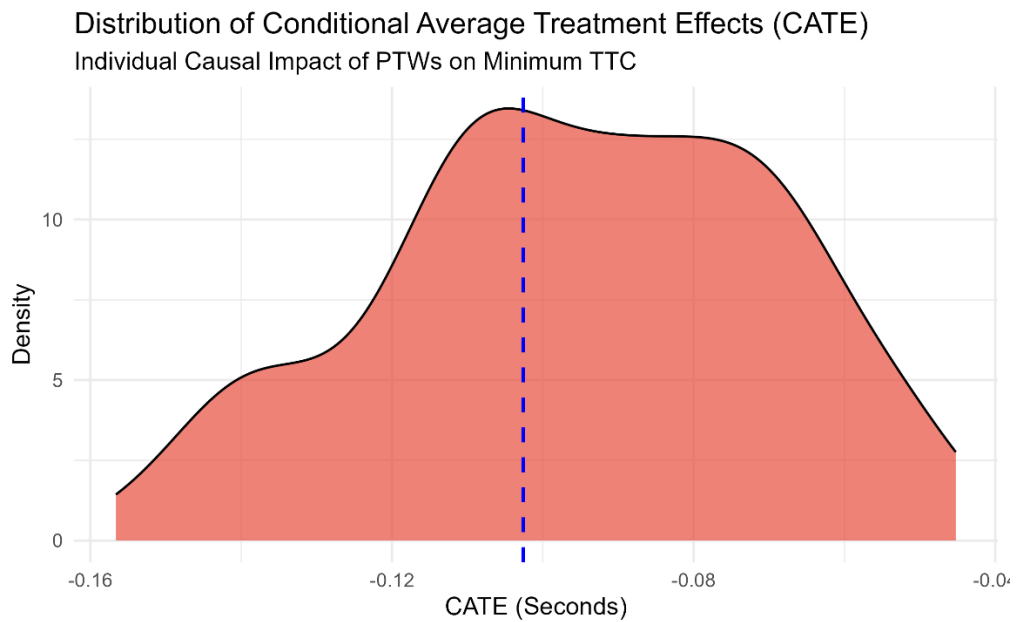


Figure 2. Distribution of Conditional Average Treatment Effects (CATE) estimated by the Causal Forest.

The scatter plot (Figure 3) confirms that in pure Car-Car interactions, drivers attempt to compensate for safety as relative speed increases; however, in motorcycle interactions, the safety margin remains dangerously close to zero across various speeds.

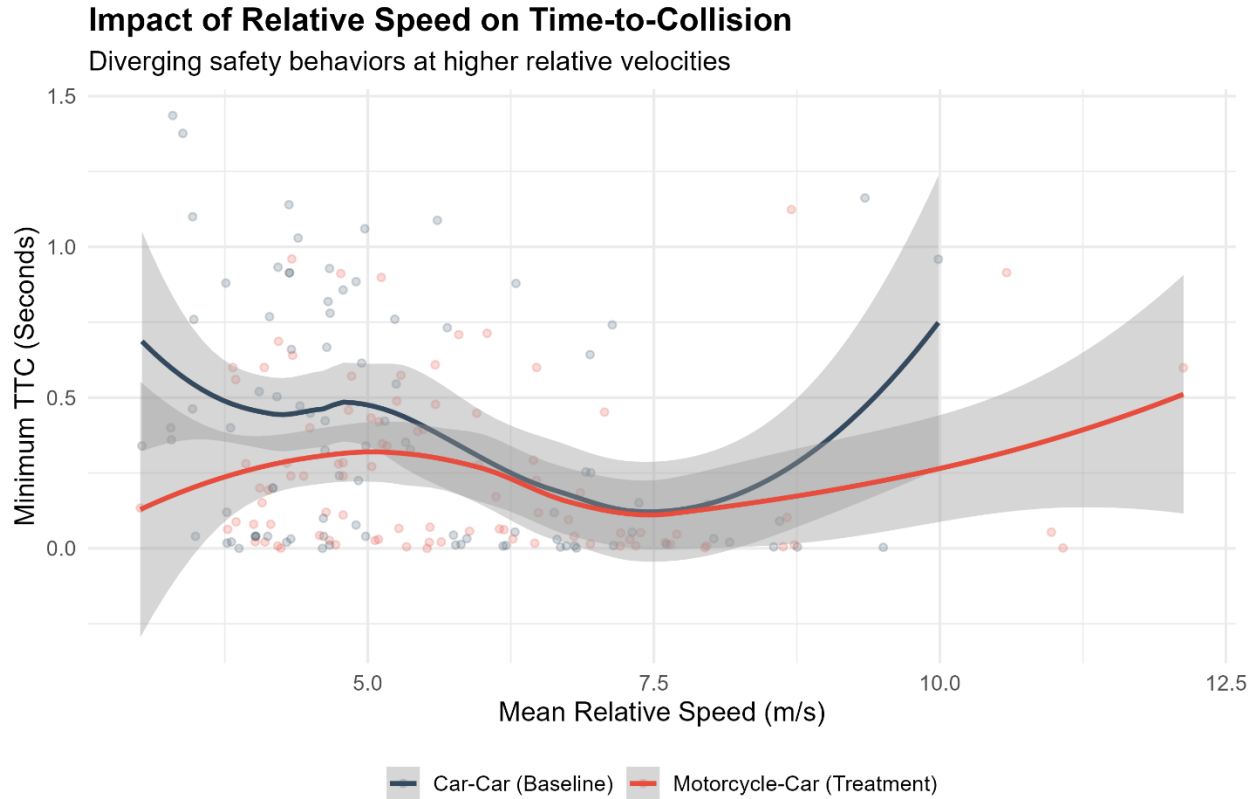


Figure 3. Impact of relative speed on Time-to-Collision degradation across interaction types.

4.2. Rosenbaum Bounds Sensitivity Analysis

To prove the model's resilience against unobserved confounders (e.g., weather conditions or driver psychological states), the Rosenbaum Bounds test was implemented. The results of the one-sided Wilcoxon test indicated that at the baseline (without hidden variables), the causal inference is statistically highly significant ($p - value = 0.0059$).

Moreover, the developed causal model is so robust that even if an unobserved variable increased the odds of exposure to the high-risk group by a factor of 3 ($\Gamma = 3.0$), the study's conclusions would retain their statistical validity and significance.

5. Conclusion

In this study, a Causal Machine Learning framework based on the Causal Forest algorithm and pNEUMA trajectory big data was developed to evaluate the isolated impact of Power-Two-Wheelers on traffic safety. By applying strict temporal filters (25-frame rolling window) and utilizing 2D Euclidean kinematics, 91 pure, noise-free near-miss events were identified.

The causal modeling findings proved that the presence of motorcycles inherently degrades the safety margin of traffic flows, causing a 0.1026-second reduction in the Time-to-Collision index, even when initial speed and distance variables are rigorously controlled. Furthermore, the overall mean TTC in the most critical events drops from 0.39 seconds for cars to 0.25 seconds for motorcycles, representing a severe reduction in available reaction time.

The high robustness of these results in the Rosenbaum Bounds sensitivity analysis confirms that the hazard posed by motorcycles is an inherent phenomenon resulting from their maneuverability and dynamic capabilities, rather than merely hidden environmental variables. These findings justify and mandate the adoption of modern engineering approaches, such as the development of segregated lanes and the revision of geometric design guidelines for mixed-traffic roadways.

References

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